Kabilan Mahathevan

Computer Science PhD Student

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Areas of Interest & Expertise

Sparse Tensor Computation - High Performance Computing - Programming Languages & Compilers - Program Analysis - Machine Learning

Education

PhD Computer Science *(Virginia Tech)* **Blacksburg VA, USA** *2025-present*

- CGPA: **4.0/4.0** & Advised by Prof. Kirshanthan ("Krish") Sundararajah.
- Working on sparse tensor compiler construction and validation.

BSc Computer Science and Engineering *(University of Moratuwa)* **Colombo, Sri Lanka** *2019-2024*

- CGPA: **3.92/4.2** (First Class)
- Dean's List & Cohort Representative for 5th and 6th Semesters.

Publications

- **Splyce: SIMD Vectorization of Sparse Coiteration**
  Kabilan Mahathevan, Poorna Gunathilaka, and Kirshanthan Sundararajah
  PACT 2026 · *Proceedings of the 35th International Conference on Parallel Architectures and Compilation Techniques*
  Winner - Chameleon Reproducibility Challenge
- **Dialect-Agnostic SQL Parsing via LLM-Based Segmentation**
  Junwen An, **Kabilan Mahathevan**, and Manuel Rigger
  SIGMOD 2026 · *Proceedings of the ACM on Management of Data*, Vol. 4, No. 3, pp. 1–28
- **TENSURE: Fuzzing Sparse Tensor Compilers (Registered Report)**
  Kabilan Mahathevan, Yining Zhang, Muhammad Ali Gulzar, and Kirshanthan Sundararajah
  FUZZING 2026 · *Proceedings of the 5th International Fuzzing Workshop* · Co-located with NDSS 2026 · doi:10.14722/fuzzing.2026.23006
- **BugsInKube: A Collection of Reconciliation Bugs**
  Kabilan Mahathevan, Sivakajan Sivaparan, Tharsha Sivapalarajah, Sunimal Rathnayake, and Ridwan Shariffdeen
  APSEC 2024 · *Proceedings of the 31st Asia-Pacific Software Engineering Conference*, pp. 447–451 · doi:10.1109/APSEC65559.2024.00057

Research Experience

PhD Student *(Virginia Tech)* **Blacksburg, USA** *09/2025 - Present*

- Optimizing sparse tensor computation to guarantee correctness and efficiency across heterogeneous hardware systems.

Visiting Research Scholar *(TEST Lab, NUS)* **Singapore** *09/2024 - 05/2025*

- Worked with Prof. Manuel Rigger in increasing the performance of the mutation testing for SQL queries and dialect-agnostic AST for SQL queries using LLM-segmentation.

Fuzzing Cloud PaaS Systems *(University of Moratuwa)* **Colombo, Sri Lanka** *07/2023 - 06/2024*

- Developed a specialized fuzzing technique to systematically uncover reconciliation bugs within Kubernetes, as a vital component of my undergraduate final year project. The empirical data on the reconciliation bug in Kubernetes has been accepted to APSEC 2024.
- Advisors: Dr. Sunimal Rathnayake & Dr. Ridwan Shariffdeen

Teaching Experience

Teaching Assistant *(Virginia Tech)* **Blacksburg, USA**

- **CS4104 - Fall 2026 - Data and Algorithm Analysis**.
- **CS1064 - Fall 2025 - Introduction to Programming in Python**.

Teaching Assistant *(University of Moratuwa)* **Colombo, Sri Lanka** *07/2023 - 06/2024*

- CS3063 - Theory of Computing, CS2043 - Operating Systems, CS3043 - Database System

Professional Experience

Software Engineer *(Sysco LABS)* **Colombo, Sri Lanka** *06/2024 - 09/2024*

- Worked on a competition organizing and judging platform from backend to frontend.
- Implemented the unit test suite, regression testing, UI functional testing suite, and the relevant CI pipeline.

Software Development Engineer Intern *(The Software Practice)* **Cecil Court, Singapore** *01/2023 - 06/2023*

- Worked on **LEAP** project for Building And Construction Authority (**BCA**), Singapore.
- Worked with the front-end team to create a better and more user-friendly interface.
- Created the full regression testing suite for the project.

Services

Artifact Reviewer

- Principles and Practice of Parallel Programming [**PPoPP'27**]
- Principles of Programming Languages [**POPL'27**]
- 23rd USENIX Conference on File and Storage Technologies [**FAST'25**]

Google Summer of Code

- Mentored SQLancer Project in GSoC 2025

Fellowships and Awards

- Winner, Chameleon Reproducibility Challenge at PACT'26 and featured in their October Newsletter.
- Student Travel Grant (PLDI-PLMW'26, VT GPSS'26, PACT'26)
- Pratt Fellowship (Virginia Tech, 2026/27)
- Mahapola Merit Scholarship (Sri Lanka, 2020-2024)

Skills

- Python
- C++
- LLVM/MLIR
- RDBMS
- Java
- Lean
- CUDA/HIP